

Mechanics 1

ID _____ Name _____

1. Select the vector values. Vector values are ()
 - a) Acceleration
 - b) Velocity
 - c) Position
 - d) Force
 - e) Time
 - f) Angle
 - g) Angular momentum
 - h) Linear momentum
 - i) Vector product
 - j) Scalar product

2. Consider the falling ball of mass m under drag force and gravity force. Set x axis to vertical direction. The drag force is proportional to the ball speed $\dot{x}$ (Coefficient is c).
 - (a) Denote the equation of motion

 - (b) Solve the equation of motion and denote the general solution. Initial condition is given as $x = \dot{x} = 0$ at time $t=0$.

3. $m(\text{kg})$ rocket is launched vertically from rest with a constant thrust $F(\text{N})$ until the rocket reached an altitude of $L(\text{m})$ and the thrust ends. Neglecting air resistance and thrust force F is larger than mg .

(a) Determine the speed of the rocket when the thrust ends.

(b) Determine the maximum height reached by the rocket.

4. Consider the movement of a particle connected to the wall by a spring. Mass of the particle is m and spring coefficient is k . Also the forced vibration $f = C \sin(\omega_0 t)$ is applied to the particle.

(a) Determine the equation of motion.

(b) Determine the general solution. Initial condition is given as $x = 0$, $\dot{x} = v_0$ at time $t=0$.